

VATSA

Video · Audio · Text · Sensory · Action

A Unified Multimodal Architecture for Human-Level Perception and Action

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Abstract

This paper introduces VATSA — a conceptual unified multimodal architecture that integrates five core perceptual and actuation streams: Video, Audio, Text, Sensory data, and Action outputs. The central proposition is that human-level intelligence arises not from mastery of any single modality, but from the seamless, context-aware fusion of all of them simultaneously.

Current state-of-the-art models — GPT-4o, Gemini Ultra, LLaVA — handle combinations of two or three modalities. VATSA proposes a five-modality unified architecture with a shared latent space, cross-modal attention, and a closed-loop action layer. This document presents the conceptual framework, motivation, proposed architecture, implementation roadmap, and intended research direction.

This is a living concept document, timestamped at conception. The author is actively developing the foundational technical skills to implement and validate each component. Contributions, collaborations, and critical feedback are welcome.

1. Motivation — Why Five Modalities?

Consider how a human doctor assesses a patient. They listen to the patient describe symptoms (Text + Audio). They observe physical appearance, movement, and non-verbal cues (Video). They read vital signs from instruments — heart rate, temperature, oxygen saturation (Sensory). And they act — prescribing, adjusting, intervening (Action). No single modality is sufficient. The diagnosis emerges from the fusion of all five.

Current AI systems are specialists. A language model understands text but cannot hear. A vision model can see but cannot respond physically. A robotics system can act but lacks deep language understanding. VATSA proposes the architectural scaffold to unify these capabilities into a single, coherent intelligence.

1.1 The Gap in Current Multimodal Research

The table below illustrates what the leading multimodal models currently cover:

Model	Video	Audio	Text	Sensory	Action	Modalities
GPT-4o (OpenAI)	Partial	Yes	Yes	No	No	3
Gemini Ultra (Google)	Yes	Yes	Yes	No	No	3
LLaVA / LLaMA 3.2	Yes	No	Yes	No	No	2
Gato (DeepMind)	Partial	No	Yes	Partial	Yes	3–4
VATSA (proposed)	Yes	Yes	Yes	Yes	Yes	5

VATSA is the only proposed architecture targeting all five modalities in a unified framework.

2. The VATSA Framework

2.1 Modality Overview

Code	Modality	Foundation Models / Techniques	Role in VATSA
V	Video	CNN, 3D-CNN, Vision Transformer (ViT), optical flow	Spatial and temporal scene understanding
A	Audio	Wav2Vec 2.0, Whisper, Mel spectrograms, RNN/LSTM	Speech, sound events, prosody, environmental audio
T	Text	Transformer, BERT, GPT, LLM fine-tuning, RAG	Language understanding, reasoning, instruction following
S	Sensory	Time-series models, IoT signal processing, LSTM	Physiological signals, environmental sensors, proprioception
A	Action	Reinforcement Learning, policy networks, diffusion models	Closed-loop physical or digital action outputs

2.2 Core Architectural Principles

VATSA is built on four foundational principles:

Principle 1: Shared Latent Space

Each modality encoder projects its input into a common high-dimensional embedding space. This allows representations from Video, Audio, Text, Sensory, and Action streams to be compared, combined, and reasoned over in a unified way. This is architecturally analogous to how CLIP aligns image and text embeddings, but extended across five modalities.

Principle 2: Cross-Modal Attention

A central cross-modal transformer attends across all five modality streams simultaneously. Rather than late fusion (combining predictions) or early fusion (concatenating raw inputs), VATSA proposes mid-fusion — each modality is independently encoded and then fused at the representation level through multi-head cross-attention. This allows each modality to influence and be influenced by the others dynamically.

Principle 3: Temporal Coherence

Video, Audio, and Sensory data are inherently temporal. VATSA must maintain temporal alignment across these streams. The architecture proposes a temporal synchronisation layer that aligns representations across modalities at matching time indices, drawing from techniques used in Audio-Visual Correspondence (AVC) learning.

Principle 4: Closed-Loop Action

Unlike pure perception models, VATSA includes an action head that produces outputs — whether physical (robotic control), digital (API calls), or communicative (language generation). The action layer is informed by all five modality streams, enabling grounded, context-aware responses. This closes the perception-action loop that is missing from most language and vision models.

3. Target Applications

The five-modality architecture makes VATSA particularly suited to environments where multiple perceptual channels carry complementary information:

3.1 Healthcare & Clinical Intelligence

- Patient monitoring: fuse vital signs (Sensory) with patient speech (Audio + Text) and physical observation (Video) to support clinical decision-making
- Surgical assistance: real-time Video + Sensory feedback combined with procedural knowledge (Text) to guide or assist robotic surgical systems
- Mental health assessment: prosody analysis (Audio) combined with language content (Text) and facial expression (Video) for holistic emotional state modelling

3.2 Industrial & Regulated Environments

- GxP pharmaceutical manufacturing: combine sensor readings (Sensory) with equipment video feeds (Video), operator instructions (Text + Audio), and automated corrective action (Action)
- Quality control: detect anomalies across multiple signal types simultaneously, with explainable outputs for audit compliance

3.3 Autonomous Systems & Robotics

- Embodied agents that perceive their environment across all five streams and respond with grounded physical actions
- Human-robot collaboration: systems that understand spoken instruction (Audio + Text), observe the human's physical context (Video), monitor shared workspace sensors (Sensory), and act accordingly (Action)

3.4 Education & Adaptive Learning

- Learner state modelling: track engagement through video (facial expression), audio (speech hesitation), and interaction patterns (Text + Action) to adapt content delivery in real-time

4. Implementation Roadmap

VATSA will be implemented iteratively. Each phase builds a working modality component, validated independently before integration. This roadmap reflects the author's current learning trajectory.

Phase	Timeline	Technical Milestone	Deliverable
1	Now – Q3 2026	CNN image classifier; object detection; transfer learning with ResNet/EfficientNet	V-module: working visual encoder with benchmark results
2	Q3 – Q4 2026	RNN, LSTM, sequence modelling; Wav2Vec fine-tuning on speech datasets	A-module: audio encoder; speech classification demo
3	Q4 2026 – Q1 2027	Transformer deep dive; fine-tune LLM on domain-specific text; RAG integration	T-module: text encoder with reasoning capability
4	Q1 – Q2 2027	Time-series modelling; IoT signal fusion; LSTM for physiological data	S-module: sensory stream encoder with synthetic health data demo
5	Q2 – Q3 2027	Cross-modal attention; shared embedding space; modality alignment	V+T fusion demo; V+A+T tri-modal prototype
6	Q3 2027 – Q1 2028	Full VATSA integration; RL-based action layer; end-to-end training	VATSA prototype + arXiv paper submission

5. Relationship to Existing Work

VATSA builds on and extends a rich body of multimodal research spanning foundational architectures (2017-2023) and the most recent unified systems (2024-2025).

5.1 Foundational Architectures

- CLIP (Radford et al., 2021) — contrastive image-text alignment; VATSA extends this shared embedding principle across five modalities rather than two
- Gato (Reed et al., 2022, DeepMind) — a generalist multi-task agent; VATSA extends this with a dedicated Sensory stream and explicit five-modality unification
- Flamingo (Alayrac et al., 2022) — visual language models with cross-attention; VATSA adopts this mechanism and generalises it across all five streams
- GPT-4o (OpenAI, 2024) — native audio-visual-text model covering three modalities; VATSA adds dedicated Sensory and closed-loop Action streams
- AudioPaLM (Google, 2023) — audio-language fusion; incorporated in VATSA as the Audio-Text integration layer within the five-modality framework

- Vaswani et al. (2017) — the Transformer architecture underpins the cross-modal attention mechanism central to VATSA's fusion layer

5.2 Recent Unified Multimodal Systems (2024-2025)

A wave of 2024-2025 research has validated the direction VATSA proposes, while revealing the gap it aims to fill:

- Uni-MoE (Li et al., 2024/2025) — a Mixture-of-Experts unified MLLM handling audio, speech, images, text, and video with modality-specific experts and cross-modality alignment. Validates VATSA Principles 1 and 2 but still lacks a dedicated Sensory stream and closed-loop Action output — precisely the gaps VATSA targets.
- ShaLa (Cui et al., 2025) — a generative framework that explicitly learns shared latent representations across modalities via inference and diffusion models. Provides strong theoretical grounding for VATSA Principle 1 and directly addresses the dynamic modality weighting question raised in Section 6.
- Zhao et al. (2025) — unified multimodal understanding and generation with shared representations and hybrid fusion strategies; supports VATSA's mid-fusion design philosophy.
- Yang et al. (2025) — a comprehensive 700+ paper survey of unified multimodal systems documenting the evolution from 2-3 modality models toward true unified architectures. Explicitly identifies the absence of Sensory and Action integration as a frontier challenge — confirming VATSA's research positioning.

5.3 Embodied AI, Action Grounding & Vision-Language-Action Models

- Ma et al. (2024) — the first dedicated survey on Vision-Language-Action (VLA) models for embodied AI; taxonomy of models grounding vision and language into physical actions. Directly supports VATSA's Action stream and Principle 4 (Closed-Loop Action).
- Liu et al. (2024/2025) — comprehensive survey on Embodied AI covering multimodal perception, sensory-action loops, sim-to-real transfer, and world model integration. Validates VATSA's robotics and healthcare applications and confirms the necessity of a dedicated Sensory stream.

Taken together, this 2024-2025 body of work confirms that the field is converging toward unified multimodal architectures — but has not yet produced a system integrating all five of VATSA's modalities. The primary contribution of VATSA is the proposed five-stream unified architecture including a dedicated Sensory channel and closed-loop Action layer that no existing system yet implements end-to-end.

6. Open Research Questions

The following questions will guide the research phase of this project. Where relevant, recent literature is cited as a starting point for investigation.

- How should modality weights be dynamically balanced when certain streams are absent or noisy? (ShaLa, Cui et al. 2025, offers a diffusion-based approach to this problem that VATSA will investigate.)
- What is the minimum viable shared embedding dimensionality to preserve information across five modalities without catastrophic forgetting? (Cui et al. 2025 provides early empirical grounding; VATSA will extend to five modalities.)
- How can temporal alignment be maintained across modalities with fundamentally different sampling rates — video at 30fps, physiological sensors at 1Hz, and text at variable intervals? (Identified as an open problem in Liu et al. 2024/2025.)
- Can VATSA be trained efficiently without large-scale compute, through curriculum learning, modular pre-training, or parameter-efficient fine-tuning (LoRA, adapters)?
- What benchmarks exist for evaluating five-modality systems, and what new benchmarks must be created? (Yang et al. 2025 survey identifies this as a critical gap in the field.)
- How should VLA-style action grounding (Ma et al. 2024) be extended to incorporate Sensory feedback as a conditioning signal, enabling truly closed-loop five-modality control?
- What are the ethical and regulatory implications of a system that simultaneously perceives video, audio, physiological sensor data, and text about a human — particularly in GxP-regulated healthcare environments?
- Can Mixture-of-Experts architectures (as demonstrated by Uni-MoE, Li et al. 2024) provide an efficient path to scaling VATSA without proportional increases in compute cost?

7. A Note on This Document

This concept paper is intentionally published at the idea stage — before implementation, before results, before peer review. It is a deliberate act of intellectual honesty and public accountability.

The author is a practitioner-researcher transitioning from eight years of AI product delivery in regulated pharmaceutical environments into full-time AI/ML research and engineering. The DBA programme at Walsh College (2025–2028) provides the research structure. The PGP at Texas McCombs School of Business (2025–2026) provides applied technical depth.

VATSA represents this author's north star — the research destination that gives direction and purpose to every CNN, RNN, LSTM, and Transformer component being learned today. Each module built along the way is not just a project. It is a component of something larger.

This document is version 1.0. It will be updated as understanding deepens, components are built, and the research community provides feedback. The arXiv preprint will follow when experimental results are available to support the claims made here.

Collaborators, researchers, and practitioners interested in multimodal AI, regulated AI systems, or the intersection of business intelligence and deep learning are warmly invited to connect.

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References

Foundational Work

Vaswani, A. et al. (2017). Attention Is All You Need. arXiv:1706.03762

Radford, A. et al. (2021). Learning Transferable Visual Models From Natural Language Supervision. OpenAI. arXiv:2103.00020

Reed, S. et al. (2022). A Generalist Agent. DeepMind. arXiv:2205.06175

Alayrac, J.B. et al. (2022). Flamingo: a Visual Language Model for Few-Shot Learning. DeepMind. arXiv:2204.14198

LeCun, Y. (2022). A Path Towards Autonomous Machine Intelligence. Meta AI.

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Barrault, L. et al. (2023). AudioPaLM: A Large Language Model That Can Speak and Listen. Google Research. arXiv:2306.12925

OpenAI. (2024). GPT-4o System Card. openai.com

Recent Unified Multimodal Architectures (2024-2025)

Li, Y. et al. (2024/2025). Uni-MoE: Scaling Unified Multimodal LLMs with Mixture of Experts. arXiv:2405.11273. Also published in IEEE Transactions on Pattern Analysis and Machine Intelligence, 2025.

Cui, J. et al. (2025). ShaLa: Multimodal Shared Latent Space Modelling. arXiv:2508.17376.

Zhao, S. et al. (2025). Unified Multimodal Understanding and Generation Models. arXiv:2505.02567.

Yang, Y. et al. (2025). A Survey of Unified Multimodal Understanding and Generation: Advances and Challenges. arXiv preprint.

Embodied AI, Action Grounding & VLA Models (2024-2025)

Ma, Y. et al. (2024). A Survey on Vision-Language-Action Models for Embodied AI. arXiv:2405.14093.

Liu, Y. et al. (2024/2025). Aligning Cyber Space with Physical World: A Comprehensive Survey on Embodied AI. arXiv:2407.06886. Published in IEEE/ASME Transactions on Mechatronics.